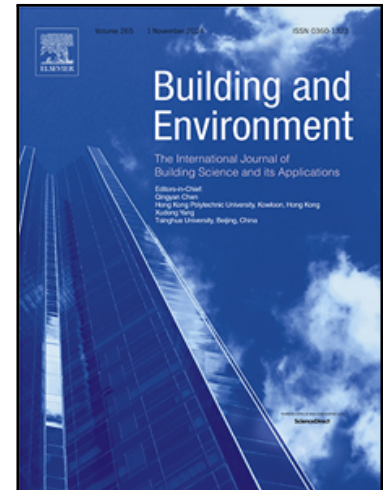


Color Matters: Investigating Stress Recovery in VR-Simulated Interior Environments Using fNIRS and Anxiety Measures

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Highlights

- Wall color significantly modulates neural and anxiety response after acute stress.
- Red and purple hues elevate both brain activation and subjective anxiety.
- White and blue walls support lower cortical activity and emotional stability.
- Recovery effects vary dynamically across early, mid, and late time intervals.
- A VR-fNIRS paradigm reveals time-sensitive color impacts on stress regulation.

Color Matters: Investigating Stress Recovery in VR-Simulated Interior Environments Using fNIRS and Anxiety Measures

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Abstract

Psychological stress is a growing global health concern, contributing to anxiety, burnout, and chronic illness. While traditional interventions focus on individual behaviors, environmental design—especially interior space where people spend most of their time—offers a scalable avenue for stress modulation. Among environmental stimuli, color is a powerful yet underutilized perceptual factor known to influence emotion and cognition. This study examines the impact of wall colors on acute stress recovery using a multimodal approach that combines immersive virtual reality (VR), functional near-infrared spectroscopy (fNIRS), and self-reported anxiety assessments. A total of 38 participants were exposed to seven monochromatic colored-wall conditions (red, blue, orange, yellow, green, purple, and white), representing distinct hues across the visible color spectrum, during a 60-second post-stress recovery period.

Brain activation in prefrontal regions and subjective anxiety levels were analyzed. Red and purple walls heightened activation in the DLPFC, PreM&SMC, and FEF, indicating increased cognitive control, motor readiness, and attentional vigilance, which aligned with elevated anxiety ($p < .05$). In contrast, white and blue reduced activation and supported emotional downregulation ($p < .05$), suggesting a restorative effect. Time-course analyses revealed early reactivity under red (0-20s) and delayed, stable responses under white (40-60s), highlighting color-specific modulation of stress-related neural functions and offering practical insights for designing stress-mitigating environments.—By integrating neural and behavioral evidence, the findings underscore the importance of color as a time-sensitive environmental variable, offering insights for the evidence-based design of stress-sensitive spaces.

Keywords: Stress Recovery, Environmental Color, Virtual Reality (VR), Near-Infrared Spectroscopy (fNIRS), Multimodal Data Collection

1. Introduction

In contemporary society, psychological stress has emerged as a pervasive and escalating global concern. The accelerated pace of urbanization, ubiquitous technological connectivity, and growing socio-economic uncertainties have significantly elevated the prevalence and intensity of stress across diverse populations [1, 2]. Psychological stress, once considered an individual challenge, is increasingly recognized as a systemic issue with profound public health implications [3]. Chronic exposure to stress has been linked to numerous adverse outcomes, including impaired immune function, cardiovascular disease, anxiety disorders, and depression [4-6]. It is now one of the leading contributors to global disease burden. Neuroscientific evidence has demonstrated that prolonged exposure to stress induces structural and functional alterations in key brain regions, particularly the prefrontal cortex, hippocampus, and amygdala [7-9]. The prefrontal cortex is primarily responsible for executive function and emotional regulation; the hippocampus supports memory consolidation and contextual processing; and the amygdala plays a central role in detecting threat and generating emotional responses, especially fear and anxiety [10]. Dysregulation within these networks can compromise both cognitive and emotional resilience, leading to diminished quality of life, productivity loss, and heightened vulnerability to mental illness [11, 12]. Given its multifactorial nature, stress must be addressed not only through individual interventions but also through broader environmental and systemic strategies.

While pharmacological treatment, psychotherapy, and behavioral interventions remain essential in managing stress at the individual level, growing attention has been directed toward the role of environmental factors in shaping psychological well-being [13]. Among

these, the built environment—specifically interior spaces—has been identified as a critical yet underleveraged domain for long-term stress regulation. People in industrialized societies typically spend over 90% of their time indoors, making the characteristics of interior spaces a constant and cumulative influence on human physiology and psychology [14]. The built environment influences mood, attention, arousal, and behavior through a variety of sensory pathways. Spatial geometry, acoustics, lighting, and material properties interact to form ambient conditions that can either aggravate or alleviate psychological strain [15, 16]. Within this context, architectural and environmental psychology converge to advocate for the design of restorative spaces—spaces that actively support mental recovery, emotional stability, and cognitive restoration. On the other hand, the Attention Restoration Theory (ART) posits that natural or restorative environments can help replenish depleted attentional resources by promoting effortless, involuntary attention [17]. However, to design such spaces effectively, a mechanistic understanding of how specific environmental features interact with neuropsychological processes is essential. A central tenet emerging from this line of inquiry is that environments are not neutral containers of activity but are active modulators of human functioning. As such, interventions embedded in the physical design of interior environments offer scalable, sustainable, and non-invasive means to mitigate chronic stress exposure and enhance everyday psychological resilience [18, 19].

Among the many environmental variables affecting psychological functioning, color holds a unique position. As one of the most immediate and universally experienced visual stimuli, color exerts potent effects on perception, attention, emotion, and physiological arousal [20, 21]. Empirical studies have consistently demonstrated that different hues elicit distinct emotional and cognitive responses [22, 23]. For instance, red is commonly associated with increased heart rate, heightened alertness, and elevated tension, while blue and green are generally linked to relaxation, reduced anxiety, and lowered physiological arousal [24, 25].

Neuroimaging studies have further shown that exposure to different colors can modulate activity in brain regions such as the dorsolateral prefrontal cortex (DLPFC), which is implicated in executive control and emotion regulation [26]. These findings suggest that color is not merely an aesthetic dimension of space but a neuroactive component with measurable psychological consequences. In addition to emotional and physiological effects, color also modulates attentional processes by altering visual salience and engaging top-down attentional control networks, particularly in the frontoparietal cortex and early visual areas [27, 28]. Despite a growing body of research linking color to emotion and cognition, the literature on how color influences recovery from acute stress remains fragmented and methodologically inconsistent. Many prior studies have relied predominantly on self-reported mood states or preference rankings, limiting their capacity to uncover the underlying neurophysiological mechanisms [29]. Furthermore, real-world studies are often confounded by environmental noise, uncontrolled lighting, and contextual variability, complicating the isolation of color effects. These limitations highlight the need for a mechanistic framework that links specific hues to neural activation in stress-regulatory circuits, and in turn to measurable recovery outcomes. Clarifying this pathway would support the application of color-based strategies in the design of psychologically restorative environments.

To advance the understanding of color's role in stress regulation, it is necessary to move beyond correlational or observational studies and adopt experimental frameworks that can isolate chromatic influences, map their temporal dynamics, and relate them to measurable neuropsychological outcomes. Virtual Reality (VR) technology offers a robust experimental platform to overcome the methodological limitations inherent in traditional environmental studies. VR allows for the creation of immersive, photorealistic environments in which specific design features—such as wall color—can be systematically manipulated while maintaining high ecological validity [30]. These environments replicate real-world settings

without the constraints of physical reconstruction, offering flexibility, replicability, and precise control over visual stimuli [31]. VR-based experiments also facilitate integration with advanced neuroimaging tools such as functional near-infrared spectroscopy (fNIRS), which measures cortical hemodynamic responses with high temporal and spatial specificity. Studying the effects of wall color on stress recovery requires both controlled visual presentation and real-time monitoring of brain responses during naturalistic interactions. VR offers a unique advantage by enabling complete control over hue, saturation, and spatial layout within immersive environments, while eliminating contextual variability and ensuring participant safety during stress induction. fNIRS, compared to fMRI and EEG, balances portability with robustness: it tolerates head movement, integrates easily with VR headsets, and captures prefrontal hemodynamic changes with sufficient temporal resolution [32, 33]. This combination allows for ecologically valid yet experimentally precise investigation of color-induced neural modulation over time. The combination of VR and fNIRS enables researchers to track real-time brain activity during exposure to environmental stimuli, offering a multidimensional view of how perceptual experiences translate into neural and psychological responses. This integration enables rigorous testing of hypotheses regarding ecological influences on stress, cognition, and affect, thereby bridging the gap between experimental control and ecological realism. Despite its potential, the combined use of VR and fNIRS to study color-based interventions for stress recovery remains relatively unexplored. A systematic, multimodal investigation of this kind could yield valuable insights for the evidence-based design of therapeutic and supportive environments.

Building on the theoretical and methodological foundations, the present study focuses on acute stress, which—among the two widely recognized forms of stress (acute and chronic)—is particularly amenable to controlled experimental paradigms and closely mirrors everyday short-term stress experiences [34]. Accordingly, this research aims to investigate

how exposure to different wall colors in immersive VR environments influences acute stress recovery at both neural and psychological levels. Specifically, this study examines the effects of seven distinct hues (blue, red, orange, yellow, green, purple, and white) on brain activation patterns and subjective anxiety levels following stress induction. These colors were selected to represent distinct hues across the visible color spectrum, providing a comprehensive basis to compare chromatic influences on acute stress recovery. Empirical evidence suggests these hues differ systematically in their psychological and physiological effects [35]. To rigorously isolate hue effects, colors were manipulated within the HSL (Hue, Saturation, Lightness) color space by holding saturation and lightness constant, thus minimizing confounding influences from brightness and chroma [36]. Acute stress was experimentally induced using a standardized cognitive stressor, after which participants were exposed to virtual rooms featuring monochromatic wall conditions. fNIRS was employed to monitor activity in prefrontal regions known to be involved in emotional regulation. While the short-form State-Trait Anxiety Inventory (STAIS-5) captured subjective anxiety levels before and after each exposure.

Given these considerations, this study investigates how variations in wall color influence acute stress recovery at both neural and psychological levels. The research focuses on whether distinct hues modulate prefrontal cortical activation and subjective anxiety following stress induction. Based on existing evidence, cool colors such as blue and green are expected to facilitate recovery by reducing cognitive load and emotional arousal, whereas warm colors such as red and orange may sustain stress-related neural activity. Through the integration of immersive VR environments and fNIRS-based monitoring, the study aims to identify time-sensitive, color-specific recovery mechanisms. The findings are intended to inform evidence-based design strategies in therapeutic, educational, and public spaces, where color may serve as an active component in promoting mental restoration.

2. Method

To operationalize the research aim—examining how wall color influences stress recovery at neural and psychological levels—a within-subject experimental design was developed around three core objectives: a) validate stress induction through a standardized cognitive load task (PASAT-C) and confirm corresponding cortical activation patterns; b) measure the effects of different wall colors on prefrontal cortex activation during the recovery phase using fNIRS; c) assess subjective anxiety outcomes after exposure to each color condition using the STAIS-5 scale. The research method outlines the experimental materials, color selection logic, VR environment setup, participant information, and the neurophysiological and behavioral data collection procedures, as well as the subsequent preprocessing and statistical analyses used to evaluate color-related recovery effects, to address these objectives.

2.1 Color selection

Seven colors (blue, red, orange, yellow, green, purple, and white) were systematically manipulated in the HSL color space. To facilitate reproducibility and verify perceptual consistency, CIE Lab* values were also reported for each color, which is perceptually intuitive but device dependent. CIE Lab* provides a device-independent, quantitative description of color in terms of lightness (L^*), redness-greenness (a^*), and yellowness-blueness (b^*), and is widely used in color calibration and environmental simulation research [37]. This dual-system reporting strikes a balance between experimental control and ecological transparency.

To address potential concerns regarding the color fidelity of VR-rendered environments, each virtual wall color was measured directly from the Oculus Quest 2 display using a calibrated spectroradiometer (JETI Specbos 1211 UV-2). This step ensured that the rendered colors closely approximated the target hues within the limits of the display gamut, allowing for post-hoc confirmation of device-based variations [38]. While some degree of perceptual shift is inevitable due to headset luminance, viewing angle, or light scattering effects, the spectrally measured values provide a rigorous reference that mitigates subjective or hardware-induced discrepancies [39]. The final color parameters, including HSL and corresponding CIE $L^*a^*b^*$ values for each condition, are summarized in Table 1. White was included as a neutral benchmark due to its prevalence in interior architectural environments and its low hue-related affective loading.

Table 1. Seven vertical wall color parameters.

Color	Sample	H	S	L	L^*	a^*	b^*
Blue		204	50	50	55.77	-7.36	-33.48
Red		356.13	50	50	45.93	51.43	23.82
Orange		25.58	50	50	56.55	23.75	41.28
Yellow		43.33	50	50	65.90	3.05	51.59
Green		130.48	50	50	68.84	-56.56	42.59
Purple		301.12	50	50	50	65.3	-40.4
White		60	100	100	100	-0.1	0.3

2.2 Data collection

The study collected two types of data to explore the changes in neurophysiological and psychological responses during stress recovery. Neurophysiological data were obtained to access brain activity variations, while psychological data assessed the impact of

environmental colors on stress levels. Multi-modal measurements comprehensively explore the mechanisms behind stress recovery in different visual contexts.

Neurophysiological data. Neurophysiological data were collected using a 67-channel fNIRS instrument (from KY-A, Wuhan YIRUIDE Group), which measured relative changes in oxygenated hemoglobin (HbO) and deoxygenated hemoglobin (HbR) concentrations, allowing for the observation of cortical activity during the stress recovery process. Based on Brodmann's areas [40], the regions of interest (ROIs) included the frontal pole area (FPA), dorsolateral prefrontal cortex (DLPFC), premotor and supplementary motor cortex (PreM & SMC), and frontal eye fields (FEF). These ROIs were selected due to their established roles in emotional regulation, executive control, motor planning, and visuospatial attention—functions that are critically involved in the modulation and recovery from acute stress. The functional ROIs are summarized in Table 2.

Table 2. Functions of regions of interest (ROIs) and analytical purpose.

ROIs	Functional Role	Analytical Purpose
DLPFC	Emotion regulation, working memory, and cognitive control [41, 42].	To test whether stress task induces activation in stress-regulatory regions.
FPA	High-level cognition, introspection, planning [43-45].	To assess involvement in high-order cognitive/emotional regulation.
PreM & SMC	Motor planning and hand control [46, 47].	To verify whether motor activity is induced by VR hand-controller use.
FEF	Visual attention and eye movement control [48, 49].	To validate activation from visual search behavior in stress task.

Psychological data. Psychological data were recorded to assess the regulatory effects of different room wall colors on psychological stress. The short-form Spielberger State-Trait Anxiety Inventory (STAIS-5) was employed [50, 51]. This validated self-report scale

includes five items that assess state anxiety (e.g., “I feel upset,” “I feel frightened,” “I feel nervous,” “I am jittery,” “I feel confused”), each rated on a 4-point Likert scale from 1 (Not at all) to 4 (Very much so). Participants completed the STAIS-5 both before and after each color exposure condition, providing insights into how environmental color influenced subjective anxiety during the recovery phase.

2.3 Virtual environment setup and experimental procedure

The virtual environment was designed using SketchUp 2019 and further refined in Unreal Engine 5 for realistic rendering. The space was configured as a fully enclosed, windowless room measuring 3 m × 3 m × 2.8 m, with a single door and a chair included to provide spatial reference cues for depth and scale [52]. All textures and materials were selected to approximate real-world indoor conditions, enhancing ecological validity [53]. Lighting was uniformly set at 300 lux.

To isolate the effect of wall color, only the wall surfaces were manipulated across conditions. The floor and ceiling remained unchanged, rendered in a neutral gray tone with low chromatic intensity. This design choice was intended to reduce visual confounds and ensure that observed psychological or neural differences could be attributed specifically to wall color variation. Walls were prioritized because, in seated indoor settings, they occupy the primary field of view and exert the most consistent influence on spatial perception and emotional experience [54]. Participants viewed the environment through an Oculus Quest 2 VR headset. Fig. 1 provides a visual overview of the experimental setup, including the virtual environment (a), participant instrumentation (b), and procedural timeline (c).

The experiment was conducted using a three-stage paradigm [55] for acute stress induction and recovery, implemented within an immersive VR environment. Participants underwent

baseline assessment, stress induction via a cognitive task, and a subsequent recovery phase in virtual rooms with colored walls.

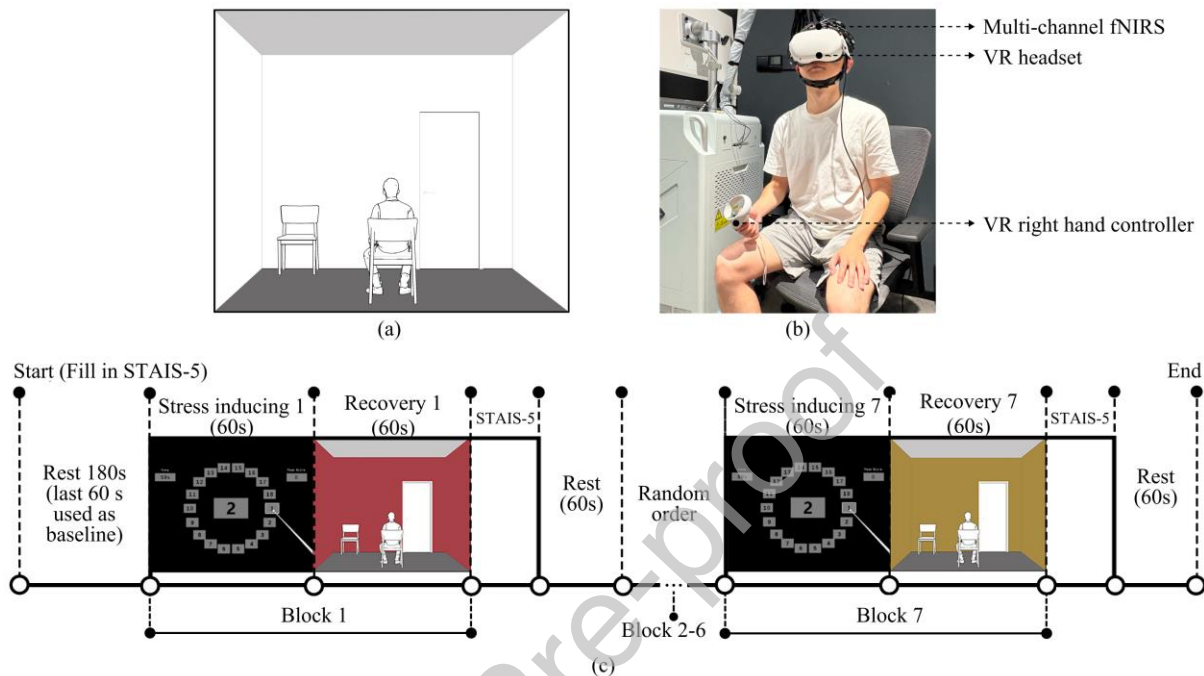


Figure 1. Overview of the virtual environment and experimental procedure. (a) VR room rendering showing a colored wall with spatial references such as a door and a chair; (b) Participant equipped with multi-channel fNIRS cap, VR headset, and right-hand controller during data acquisition; (c) Experimental procedure including baseline rest, PASAT-C stress task, colored-room recovery, and inter-block dark rest.

At the start of each session, participants completed the STAIS-5 to establish a baseline measure of state anxiety. Following this, they were equipped with an fNIRS cap and an Oculus Quest 2 VR headset. After a 180-second resting period in a dark virtual environment, the final 60 seconds were recorded as the neurophysiological baseline. The core experimental task consisted of seven blocks, each including a 60-second mental arithmetic challenge based on the Paced Auditory Serial Addition Task – Computerized (PASAT-C), widely used for acute stress induction [56, 57]. During this task, a random single-digit number (ranging from

1 to 9) appeared on the VR screen, and participants were required to compute and select the correct sum of the most recently presented number and the one immediately preceding it. The interval between successive numbers was fixed at 2.7 seconds, as this timing was identified during pilot testing to provide an optimal balance between task difficulty and feasibility. The VR interface was designed with a standard gray (L=50) color scheme on a black background to minimize potential color-induced cognitive biases. To further increase performance pressure, participants were able to view a countdown timer and their real-time score throughout the task. Immediately following each stress task, participants were transitioned into a virtual room featuring a single-color wall. The colors were randomly ordered across blocks to counterbalance sequence effects. A 60-second recovery window was selected because prior study shows that acute physiological and neural stress indicators can exhibit measurable recovery toward baseline within short periods (approximately 60s) after stress cessation [58], while a longer interval would unnecessarily extend the total session and increase participant fatigue. During the 60-second recovery period, participants were instructed to remain still, relax, and clear their minds. fNIRS continuously recorded cortical activation throughout this interval. Upon completion of each recovery phase, the STAIS-5 was re-administered to capture changes in subjective anxiety. To minimize carry-over effects, each block concluded with a 60-second dark rest interval before the subsequent trial.

2.4 Experimental environment and participants

The experiment was conducted in a dark and quiet psychological laboratory designed to minimize external disturbances and ensure a controlled setting for accurate data collection.

The sample size for this study was determined using G*Power 3.1.9.7, which revealed a minimum required sample size of 21 participants. These calculations were based on an effect

size of 0.25 (medium effect) [59], a significance level of 0.05, and a power of 0.80, ensuring that the study meets the required sample size criteria.

Participants ($N = 38$; 20 males, 18 females; age 21–33) were recruited through online campus advertisements. All were right-handed university students with normal color vision, no neurological or psychiatric history, and prior experience with VR. Before the experiment, participants completed a short VR adaptation session and were verbally asked whether they experienced any symptoms of discomfort; none reported significant issues. During the experiment, participants remained seated and were instructed to minimize head movement to reduce fNIRS artifacts. The study was approved by the university ethics committee (HKUST(GZ)-HSP-2024-0017), and informed consent was obtained.

2.5 Data pre-processing and statistical analysis

The fNIRS data were pre-processed using the NIRS-KIT toolbox [60] in MATLAB 2014b. To prepare the fNIRS signals for analysis, several preprocessing steps were applied sequentially. Slow signal drifts were first removed through linear detrending by fitting and subtracting a low-order polynomial from each channel's time series. Motion artifacts were then corrected using the Temporal Derivative Distribution Repair (TDDR) algorithm, followed by a third-order Butterworth band-pass filter with cut-off frequencies of 0.01 Hz and 0.2 Hz to eliminate physiological and high-frequency noise [61]. After these corrections, the optical density data were converted into changes in oxygenated (HbO) and deoxygenated hemoglobin (HbR) concentrations using the modified Beer–Lambert law [62]. As HbO signals generally exhibit a higher signal-to-noise ratio and greater sensitivity to task-related changes, subsequent analyses focused exclusively on HbO [63].

The fNIRS analysis was conducted at the ROI level [64]. Paired-samples t-tests compared HbO concentrations between baseline and stress phases to identify stress-responsive regions. These ROIs were then included in two-way RM-ANOVAs (Color $\times$ Time) to examine recovery-phase effects. For STAIS-5 scores, paired-samples t-tests compared each color condition with baseline, followed by a one-way RM-ANOVA across conditions. Pearson correlation analyses were conducted between STAIS-5 scores and HbO values from ROIs that showed significant effects during the recovery phase, across each color condition.

3. Result

The results present the neurophysiological (fNIRS) and psychological (STAIS-5) findings regarding the effects of wall color on acute stress recovery, which begin with the validation of stress induction using cortical activation during the stress task, followed by analyses of color main effects on cortical activation. Next, the interaction between color and recovery time is examined to reveal dynamic neural patterns. Subsequently, subjective anxiety ratings under each color condition are compared with baseline and across colors. Finally, the associations between cortical responses and anxiety scores are explored to assess the alignment between neural and subjective responses.

3.1 Validation of stress induction

The effectiveness of the stress-induction procedure was evaluated by testing whether the task elicited significant cortical activation. HbO signals were extracted from each ROIs, and paired-sample t-tests were conducted to compare the stress and baseline phases. The analysis focused on brain regions commonly implicated in emotion regulation and cognitive control

(i.e., DLPFC and FPA), as well as regions involved in task-related activation (i.e., FEF and PreM&SMC). As shown in Table 3, all four ROIs exhibited significantly increased activation during the stress phase compared to baseline. Specifically, activation increased in the DLPFC ($t(9) = 4.83, p < .001$), FPA ($t(9) = 3.92, p < .001$), PreM&SMC ($t(9) = 3.44, p < .01$), and FEF ($t(9) = 4.29, p < .001$).

Table 3. Brain regions significantly activated during stress induction.

ROIs	M (Stress)	SD (Stress)	M (Baseline)	SD (Baseline)	t	p	Significance
DLPFC	0.05	0.19	-0.31	0.35	4.83	< .001	***
FPA	0.04	0.24	-0.29	0.39	3.92	< .001	***
PreM&SMC	0.08	0.23	-0.25	0.46	3.44	< .01	**
FEF	0.09	0.30	-0.41	0.52	4.29	< .001	***

Note: The p value was corrected using FDR method, **: Significant at $\alpha = 0.01$ level. ***: Significant at $\alpha = 0.001$ level.

These results indicate that the stress task successfully induced acute stress, eliciting significant cortical activation in regions involved in executive control (DLPFC), emotional regulation (FPA), motor planning (PreM&SMC), and attentional orienting (FEF). The spatial patterns of activation are visualized in Fig. 2.

In addition to group-level results, individual-level activation patterns were also examined. While some inter-individual variability was observed, most participants showed clear task-related hemodynamic responses in at least one ROI, supporting the general effectiveness of the stress induction. No participants were identified as non-responders, and none were excluded from the analysis.

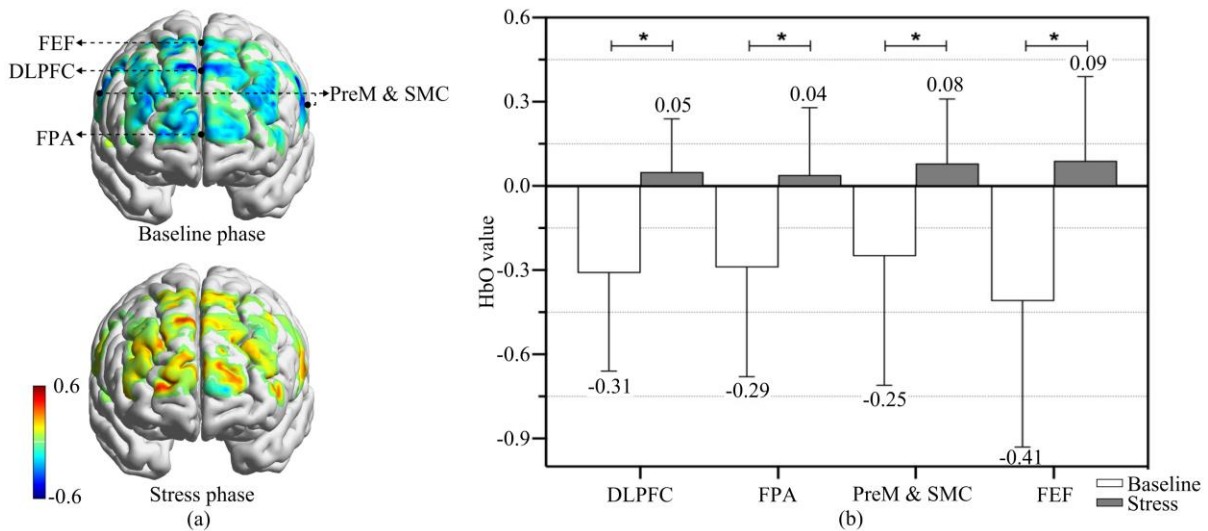


Figure 2. Validation of stress induction via cortical activation across four ROIs. (a) Group-level HbO activation maps during baseline and stress phases, showing increased activation under stress; (b) Mean HbO values in four ROIs (DLPFC, FPA, PreM&SMC, FEF) during baseline (white bars) and stress (grey bars). Error bars indicate SEM. Asterisks denote FDR-corrected significance: $p < .05$, $p < .01$, $p < .001$.

3.2 Neurophysiological response

Building upon the confirmed activation of brain regions during acute stress induction, the following analyses investigated how wall color modulated neural recovery dynamics. Cortical activation was assessed across different color exposures and over sequential recovery phases. Repeated measures analyses of variance (RM-ANOVAs) were performed with Color and Time as within-subject factors, with corrections for multiple comparisons. The findings are structured to present the main effects of color, the temporal evolution of activation, and the color-by-time interaction effects.

3.2.1 Effect of wall color on cortical activation

The influence of wall color on cortical activation was first examined by assessing the main effects of Color across predefined ROIs. A repeated measures ANOVA revealed a significant main effect of Color in all three regions: DLPFC ($F(6, 216) = 2.52, p < .05, \eta^2 = 0.07$), PreM&SMC ($F(6, 216) = 5.56, p < .001, \eta^2 = 0.13$), and FEF ($F(6, 216) = 2.74, p < .01, \eta^2 = 0.07$).

Post hoc comparisons with FDR correction showed that in the DLPFC, the red condition ($M = 0.06, SD = 0.06$) showed significantly greater activation than white ($M = -0.15, p < .05$). Similarly, yellow ($M = 0.06, p < .05$) and purple ($M = 0.05, p < .05$) also showed greater activation than white.

In the PreM&SMC, the red condition ($M = 0.21, SD = 0.07$) showed the highest activation, significantly greater than the orange, yellow, purple, and white conditions ($p < .05, .05, .01$, and $< .001$, respectively). In addition, the blue, orange, yellow, and green conditions all showed significantly greater activation than the white condition ($p < .05$ for all). Finally, the green condition showed significantly greater activation than the purple condition ($p < .05$).

In the FEF, the red condition ($M = 0.08$) showed significantly greater activation than both green ($M = -0.18, p < .05$) and white ($M = -0.15, p < .05$).

Table 4 summarizes the statistical outcomes of the color main effect analysis and the significant pairwise comparisons based on FDR-corrected post hoc tests. Fig. 3 provides a comprehensive visualization of these effects. These visualizations highlight a consistent pattern of elevated activation under the red condition across DLPFC, PreM&SMC, and FEF, with the white condition consistently eliciting the lowest activation.

Table 4. Main effect of wall color on cortical activation at ROIs.

Main Effect	ROIs	F	p	η^2	Significant Pairwise Comparison
Color	DLPFC	2.52	< .05	0.07	Red/ Yellow/ Purple/ >White
Color	PreM&SMC	5.56	< .001	0.13	Red> Orange/ Yellow/ Purple/ White; Blue/ Orange/ Yellow/ Green> White; Green> Purple
Color	FEF	2.74	< .01	0.07	Red >Green/ White

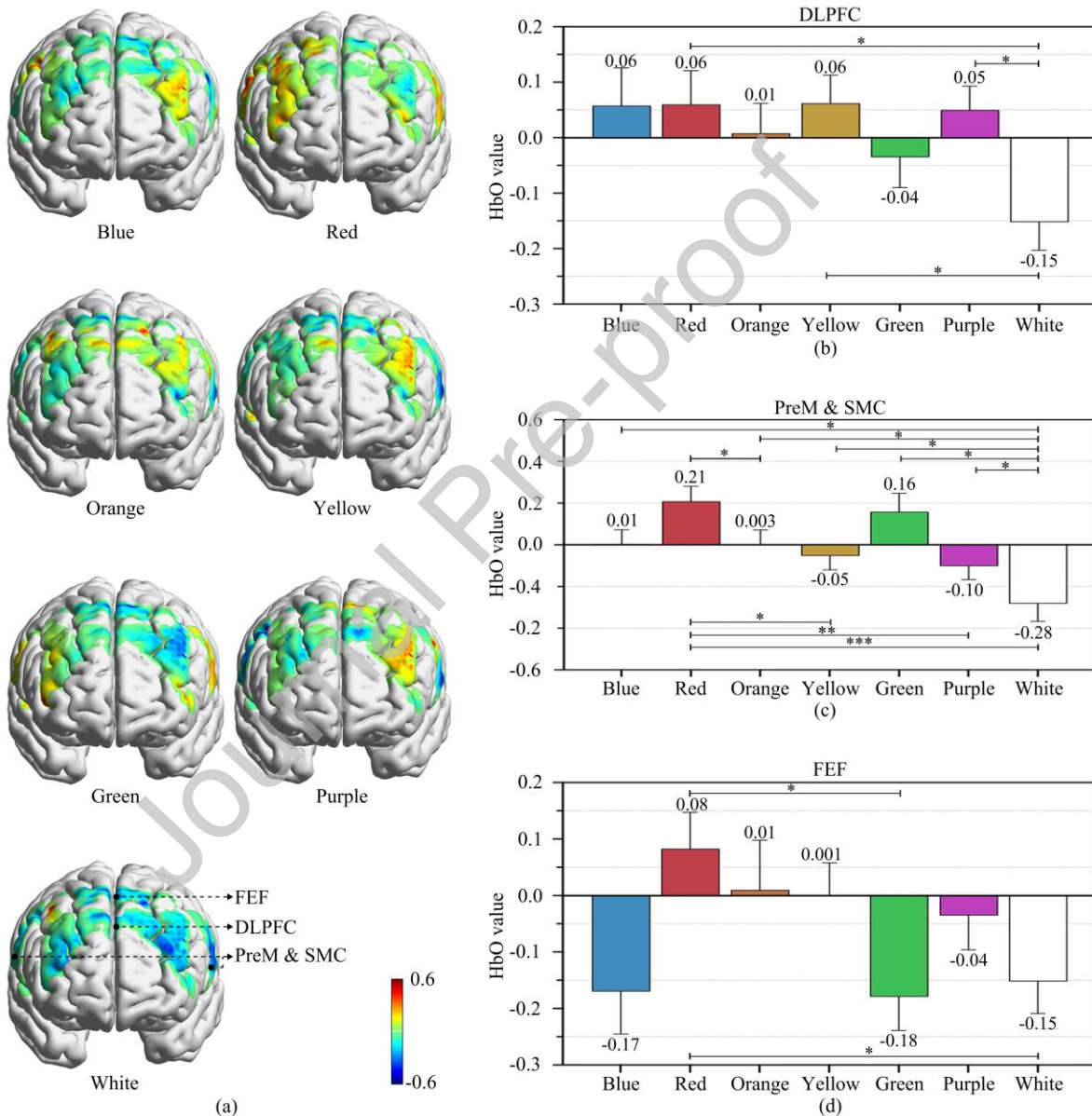


Figure 3. Main effects of wall color on cortical activation across ROIs during stress recovery. (a) Group-averaged cortical surface maps of HbO concentration for each color condition; (b–d) Mean HbO levels across DLPFC (b), PreM&SMC (c), and FEF (d) under seven color conditions. Error bars represent SEM. Asterisks indicate FDR-corrected significance.

3.2.2 Temporal dynamics of stress recovery

The temporal evolution of cortical activation during the recovery period was examined using Time as a within-subject factor. To capture dynamic changes in neural activity, the 60-second recovery period following stress induction was segmented into three consecutive intervals: T1 (0–20 seconds), T2 (20–40 seconds), and T3 (40–60 seconds). This subdivision enabled finer-grained examination of early, middle, and late recovery phases, and aligns with previous fNIRS stress research that employs short contiguous windows to capture transient post-stress neural changes [65].

A significant main effect of Time was observed in the PreM&SMC ($F(2, 72) = 4.70, p < .05, \eta^2 = 0.13$). Post hoc analyses with FDR correction revealed that activation during the second interval (T2, $M = 0.29$) was significantly greater than both the first (T1, $M = -0.12, p < .05$) and the final (T3, $M = -0.19, p < .05$) intervals.

Table 5 summarizes the statistical outcomes of the time main effect analysis and the significant pairwise comparisons based on FDR-corrected post hoc tests. Fig. 4 provides a comprehensive visualization of these effects. These results reflect a distinct temporal pattern in cortical recovery, with a mid-phase activation peak.

Table 5. Effects of recovery time segments on brain cortical activation.

Main Effect	ROI	F	p	η^2	Significant Pairwise Comparison
Time	PreM&SMC	4.70	< .05	0.13	T2> T1; T2>T3

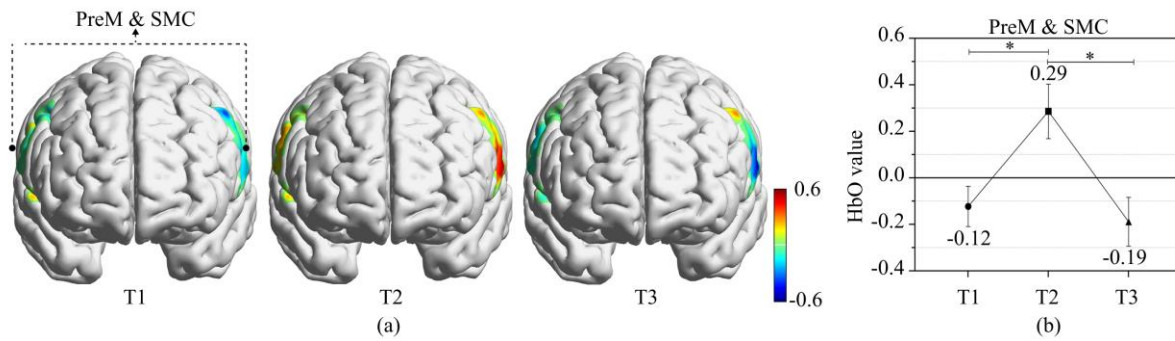


Figure 4. Temporal dynamics of cortical activation during the recovery phase. (a) Surface maps of PreM&SMC HbO activation during three consecutive 20-second recovery intervals (T1–T3); (b) Mean HbO values in the PreM&SMC region during each time interval. A mid-recovery peak in activation is observed during T2. Error bars represent SEM. Asterisks indicate statistically significant differences ($p < .05$, FDR-corrected).

3.2.3 Interaction between wall color and recovery time

To examine whether the modulation of cortical activation by wall color varied across different recovery phases, a two-way repeated measures ANOVA was conducted with Color (seven levels) and Time (three levels) as within-subject factors.

A significant Color $\times$ Time interaction effect was identified in both the DLPFC ($F(12, 432) = 1.90$, $p < .05$, $\eta^2 = 0.05$) and the FEF ($F(12, 432) = 1.81$, $p < .05$, $\eta^2 = 0.05$). Simple effects analyses with FDR correction revealed. In the DLPFC, during the mid-recovery phase (T2), the yellow condition ($M = 0.31$) showed significantly greater activation than white ($M = -0.17$, $p < .05$). In the FEF, during the early recovery phase (T1), the red condition ($M = 0.40$) showed significantly greater activation than blue ($M = -0.50$, $p < .05$), green ($M = -0.54$, $p < .01$), and white ($M = -0.59$, $p < .01$). During the late recovery phase (T3), white ($M = 0.42$) showed significantly higher activation than purple ($M = -0.23$, $p < .05$).

Table 6 summarizes the statistical outcomes of the color $\times$ time interaction effects and the corresponding pairwise comparisons following FDR-corrected post hoc tests. Fig. 5 provides

a comprehensive visualization of these effects. These findings suggest that the impact of wall color on stress recovery is temporally dynamic, with distinct color effects emerging at different recovery phases and across brain regions.

Table 6. Interaction effect of wall color and recovery time on brain activation.

Interaction Effect	ROIs	F	p	η^2	Significant Pairwise Comparison
Color $\times$ Time	DLPFC	1.9	< .05	0.05	T2: Yellow > White
Color $\times$ Time	FEF	1.81	< .05	0.05	T1: Red > Blue/ Green/ White; T3: White > Purple

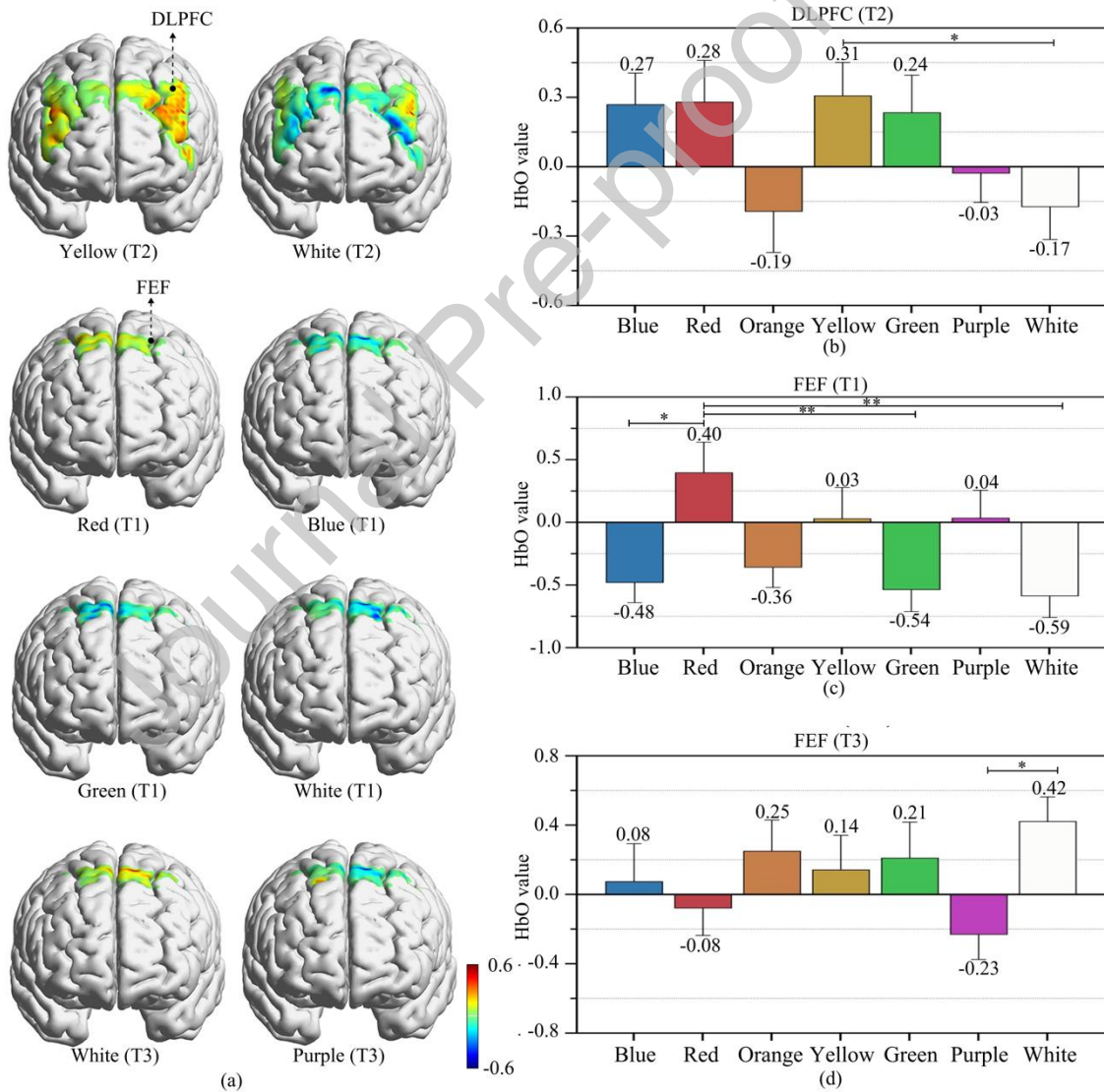


Figure 5. Interaction effects between wall color and recovery time. (a) Activation maps showing representative DLPFC and FEF responses across recovery phases under selected

colors.; (b – d) Bar plots showing mean HbO levels at specific time points: (b) DLPFC activation at T2 (yellow > white), (c) FEF activation at T1 (red > blue/green/white), (d) FEF activation at T3 (white > purple). Error bars = SEM. Asterisks denote FDR-corrected significance.

3.3 Psychological Response

Participants' subjective anxiety responses were assessed using the STAIS-5 to evaluate the perceived effectiveness of stress recovery across wall color conditions. Measurements were taken at two time points: before any task or stimulus exposure (baseline), and after the recovery period in each color condition. The baseline score reflected the unstimulated psychological state and served as the reference for evaluating recovery outcomes.

Descriptive statistics for each color condition, along with results from paired-sample t-tests comparing post-recovery scores to the baseline, are presented in Table 7. Statistical comparisons revealed that red ($t(37) = 2.92, p < .05$) and purple ($t(37) = 2.88, p < .05$) significantly increased anxiety relative to baseline. No significant differences were observed for the blue, orange, yellow, green or white conditions (all $p > .05$).

Table 7. Descriptive statistics and paired t-test results comparing post-recovery STAIS-5 scores with baseline.

Color Condition	M (Post)	SD (Post)	M (Baseline)	SD (Baseline)	t	p	Significance
Blue	6.66	2.13	6.82	2.43	-0.31	0.76	
Red	9.05	4.04	6.82	2.43	2.92	< .05	*
Orange	7.97	2.68	6.82	2.43	1.83	0.08	
Yellow	6.97	2.68	6.82	2.43	0.26	0.80	
Green	7.21	3.38	6.82	2.43	0.58	0.57	
Purple	8.66	3.42	6.82	2.43	2.88	< .05	*

White	6.84	1.70	6.82	2.43	0.05	0.96
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Note: The p value was corrected using FDR method, *: Significant at $\alpha = 0.05$ level.

These results are visualized in Fig. 6, which shows the mean STAIS-5 scores under each color condition and the baseline.

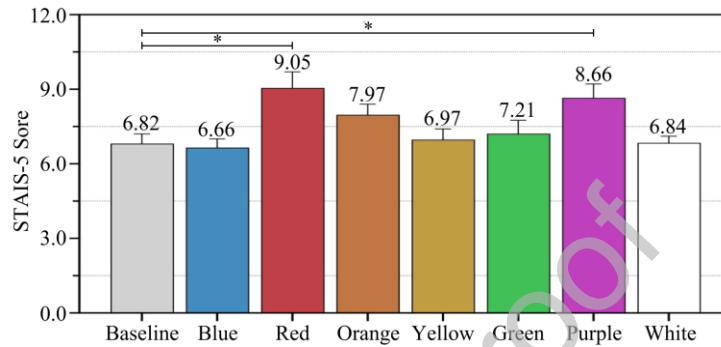


Figure 6. STAIS-5 anxiety scores across color conditions. Mean post-recovery STAIS-5 scores compared to baseline under each wall color condition. Red and purple significantly increased anxiety. Error bars indicate SEM. $p < .05$, FDR-corrected.

To examine overall differences between color conditions, a repeated measures ANOVA was performed. A significant main effect of Color was found ($F(6, 222) = 6.92$, $p < .001$, $\eta^2 = 0.16$). Post hoc comparisons with FDR correction indicated that the red condition ($M = 9.05$, $SD = 4.04$) were significantly higher than in the yellow ($M = 6.97$, $p < .01$), blue ($M = 6.66$, $p < .01$), and white ($M = 6.84$, $p < .01$) conditions. Similarly, anxiety scores in the purple condition ($M = 8.66$) were significantly higher than in the blue ($p < .05$) and white ($p < .05$) conditions. Detailed results of the ANOVA and pairwise comparisons are presented in Table 8.

Table 8. RM-ANOVA and post hoc comparisons of STAIS-5 scores by color condition.

Main Effect	F	p	η^2	Significant Pairwise Comparison
Color	6.91	< .001	0.16	Red>Yellow/ Blue/ White; Purple>Blue/ White

3.4 Correlation between neurophysiological and psychological response

To examine the convergence between neurophysiological and psychological responses during stress recovery, Pearson correlation analyses were conducted between ROI-level cortical activation and post-recovery anxiety scores (STAIS-5) under each of the seven color conditions. ROIs included the DLPFC, FEF, and PreM&SMC, which showed significant involvement in stress-related processing and recovery dynamics in earlier analyses.

As summarized in Table 9, several correlations reached statistical significance ($p < .05$). Specifically, in the yellow condition, DLPFC activation negatively correlated with anxiety ($r = -.35$); in the white condition, PreM&SMC activation positively correlated with anxiety ($r = .33$); and in the red condition, FEF activation positively correlated with STAIS-5 scores ($r = .37$). These associations are visualized in Fig. 7, which illustrates the directional trends between brain activation and self-reported anxiety under representative color conditions.

Table 9. Correlations between ROI-specific HbO value and post-recovery STAIS-5 scores across wall color conditions

ROIs	Color	r	p	Significance
DLPFC	Yellow	-0.35	< .05	*
PreM&SMC	White	0.33	< .05	*
FEF	Red	0.37	< .05	*

Note: Pearson correlation coefficients reflect the associations between HbO values in each ROI and STAIS-5 anxiety scores under individual color conditions ($N = 38$).

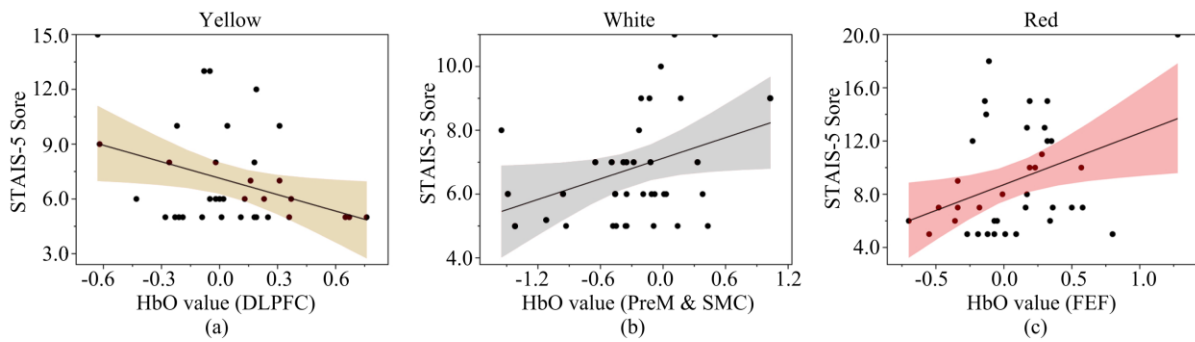


Figure 7. Correlations between cortical activation and anxiety under selected colors. Scatter plots showing ROI-specific HbO values plotted against STAIS-5 scores in: (a) DLPFC under yellow (negative correlation), (b) PreM&SMC under white (positive correlation), (c) FEF under red (positive correlation). Each point represents one participant; shaded bands indicate 95% CI. Trend-level findings, uncorrected for multiple comparisons.

However, none of the observed effects survived FDR correction for multiple comparisons and should therefore be interpreted as exploratory findings. These trend-level associations suggest possible color-specific neural correlates of subjective stress, highlighting the value of integrated neurophysiological and behavioral measures in evaluating recovery outcomes.

Table 10 summarizes the combined effects of color on brain activation, temporal recovery dynamics, and subjective anxiety. Each color condition is represented by its associated activation in brain regions, the observed time window of peak cortical response, and whether it was followed by reduced anxiety. Notably, the red condition showed early activation in the FEF, and significantly increased anxiety scores compared to baseline. The purple condition was similarly associated with elevated anxiety, though no distinct neural activation or timing pattern was observed. The yellow condition exhibited peak activation in the DLPFC during the mid-recovery phase (T2), but was not associated with significant changes in subjective anxiety. The white condition showed late-phase (T3) activation in the FEF, without a corresponding change in anxiety levels. The blue, orange, and green conditions did not produce consistent patterns in neural activation or subjective anxiety.

Table 10. Summary of color-based recovery effects on cortical activation, temporal dynamics, and subjective anxiety.

Color	Activated ROI	Time Peak	Anxiety Effect
Blue	-	-	-
Red	FEF	T1	↑
Orange	-	-	-
Yellow	DLPFC	T2	-
Green	-	-	-
Purple	-	-	↑
White	FEF	T3	-

Note: This table summarizes recovery-related effects across seven wall color conditions by integrating: (1) the activated brain region(s) during the recovery phase; (2) the time window of peak neural response (T1: 0–20s, T2: 20–40s, T3: 40–60s); and (3) anxiety outcomes based on post-recovery STAIS-5 scores (↑: increased anxiety, ↓: reduced anxiety). Data are derived from sections 3.2 and 3.3 of the results.

These results reflect divergent patterns across colors, with some conditions showing alignment between brain and behavioral responses, while others suggest a mismatch. This integrated summary provides a concise overview of color-specific recovery profiles across physiological and psychological dimensions.

4. Discussion

This study investigated how colored wall environments influence acute stress recovery at both neural and psychological levels, using VR immersion and fNIRS measurement. The findings demonstrate that wall color significantly modulates recovery-related brain activity, particularly in the prefrontal cortex, and alters subjective anxiety outcomes. Importantly, these effects were not static but varied dynamically over time, suggesting that color perception and stress regulation operate on intertwined temporal and neural trajectories.

Red environments elicited heightened activation in the DLPFC, PreM&SMC, and FEF regions (Fig. 3b–d), alongside significantly elevated post-recovery anxiety (Fig. 6). These

results align with previous findings associating warm hues with arousal and cognitive load [66], indicating that such environments may sustain or even amplify stress-related cognitive and emotional demands. In contrast, white, despite its perceptual neutrality, consistently yielded the lowest cortical activation and was associated with stable or reduced anxiety, positioning it as a favorable color for passive recovery contexts.

Compared to prior studies that primarily focused on aesthetic or affective responses to color [67, 68], this work integrates findings through the lens of neuropsychological mechanisms such as ART, and cortical emotional regulation. For example, ART posits that environments that engage involuntary attention can replenish depleted cognitive resources. The present findings extend this theory by showing that perceptual elements like hue—especially low-arousal colors such as blue and white—may facilitate such restorative attention, reflected in decreased prefrontal and motor cortex activation [69]. Similarly, the early activation observed in the frontal eye fields under red may reflect enhanced visual salience, consistent with visual processing theories that associate red with alerting mechanisms and threat detection [70, 71].

Temporal analyses revealed that stress recovery is not a uniform process but features phase-specific cortical dynamics [72, 73]. Notably, the PreM&SMC exhibited peak activation during the mid-recovery stage (T2, 20 – 40 seconds), as shown in Figure 4b. Interaction effects further highlighted the color-specific nature of recovery timing: yellow significantly increased DLPFC activation during the T2 period. At the same time, red elicited early FEF activation during T1, and white produced late FEF activation during T3 (Fig. 5b–d). These dynamic responses suggest that different colors influence recovery at different stages, rather than uniformly across time. This offers a promising theoretical framework: color-induced

attentional and emotional effects may follow a temporal arc, wherein high-salience hues exert early influence, and low-arousal hues modulate sustained regulation.

These findings offer broader insights into how environmental color functions not only as a visual attribute but as a neurophysiologically active component of spatial experience. Theoretically, the study contributes to the field of neuroarchitecture by linking chromatic variation with specific patterns of cortical engagement over time, revealing how stress-regulatory systems dynamically respond to hue. Practically, the results highlight the potential of color as a low-cost, scalable design variable for promoting psychological recovery in everyday settings—from hospitals and clinics to classrooms and workplaces. Cool and neutral hues, such as blue and white, emerged as favorable in downregulating stress-related brain activity, while warm hues sustained cognitive and emotional arousal. Importantly, these effects unfolded across distinct recovery phases, suggesting that color design may benefit from temporal adaptation rather than static application.

The exploratory correlations between neural responses and anxiety ratings (Fig. 7) provide additional insight into individual variability. Although preliminary, these associations suggest that specific cortical regions may serve as biomarkers for real-time stress regulation under different color exposures, paving the way for personalized environmental interventions. They also underscore the need for adaptive systems capable of tailoring environmental feedback (e.g., hue shifts or brightness adjustment) in response to user state.

Taken together, the current findings contribute to both theory and practice. Theoretically, they underscore the interaction between perceptual stimuli and temporal neurodynamic in emotional regulation. Practically, they offer evidence-based directions for designing dynamic recovery environments in healthcare, workplaces, or virtual platforms—environments that are not just color-tuned but time-sensitive in their application.

5. Conclusion

This study demonstrates that immersive color environments have a significant influence on acute stress recovery, both at the cortical level and in subjective experience. Specifically, warm hues such as red and purple elicited heightened prefrontal and attentional activation and were associated with elevated anxiety, whereas neutral and cool hues, like white and blue, promoted lower activation and more favorable emotional states.

These findings contribute to the emerging discourse in neuroarchitecture and environmental psychology by revealing how subtle environmental cues—like hue—can exert measurable neurophysiological effects. By linking chromatic input with recovery-phase brain activation, the study expands current knowledge on how built environments can actively shape mental states, beyond static aesthetics or occupant preferences.

The recovery process was also temporally structured, with mid-recovery intervals (20 – 40s) showing peak neural engagement, and color effects varying across these time windows. This temporal modulation suggests that future interior design might benefit from dynamic or adaptive color strategies, where spatial chromatics respond to time-of-day, user stress level, or functional zone.

These findings offer a foundation for evidence-based design strategies in therapeutic, workplace, or educational environments. For example, in hospitals or clinics, exposure to blue or white environments during initial recovery may promote down-regulation of stress-related cortical activity, while subtle yellow accents might support cognitive stability in later recovery stages. In offices, dynamic lighting schemes that shift hue intensity across the day may be used to support attention and relaxation cycles.

Beyond static color selection, the results suggest the feasibility of temporally sequenced color interventions—for instance, initiating recovery with calming hues (e.g., white or blue) and transitioning to stabilizing hues (e.g., yellow) to sustain cognitive and emotional regulation. This dynamic approach may be particularly valuable in digital well-being applications, stress-relief rooms, or adaptive VR environments.

Nevertheless, several limitations should be noted. The study employed a fixed recovery duration and limited the exposure time per color to 60 seconds, which may not capture long-term or cumulative effects. The sample size, while adequate for a within-subject design, may limit generalizability. Additionally, only fNIRS and self-report measures were used; incorporating physiological signals (e.g., heart rate, skin conductance), real-world behavioral markers, or multi-sensory stimuli would provide a richer understanding of environmental recovery. Future research should explore longer and adaptive exposure sequences, validate color timing strategies in real-world or clinical settings, and consider individual differences in color-emotion associations. Extending this work to architectural prototypes or full-scale built environments would help bridge the gap between simulation and application.

Ultimately, this work highlights the importance of designing for time, not just space, in the application of color-based interventions for mental well-being.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.